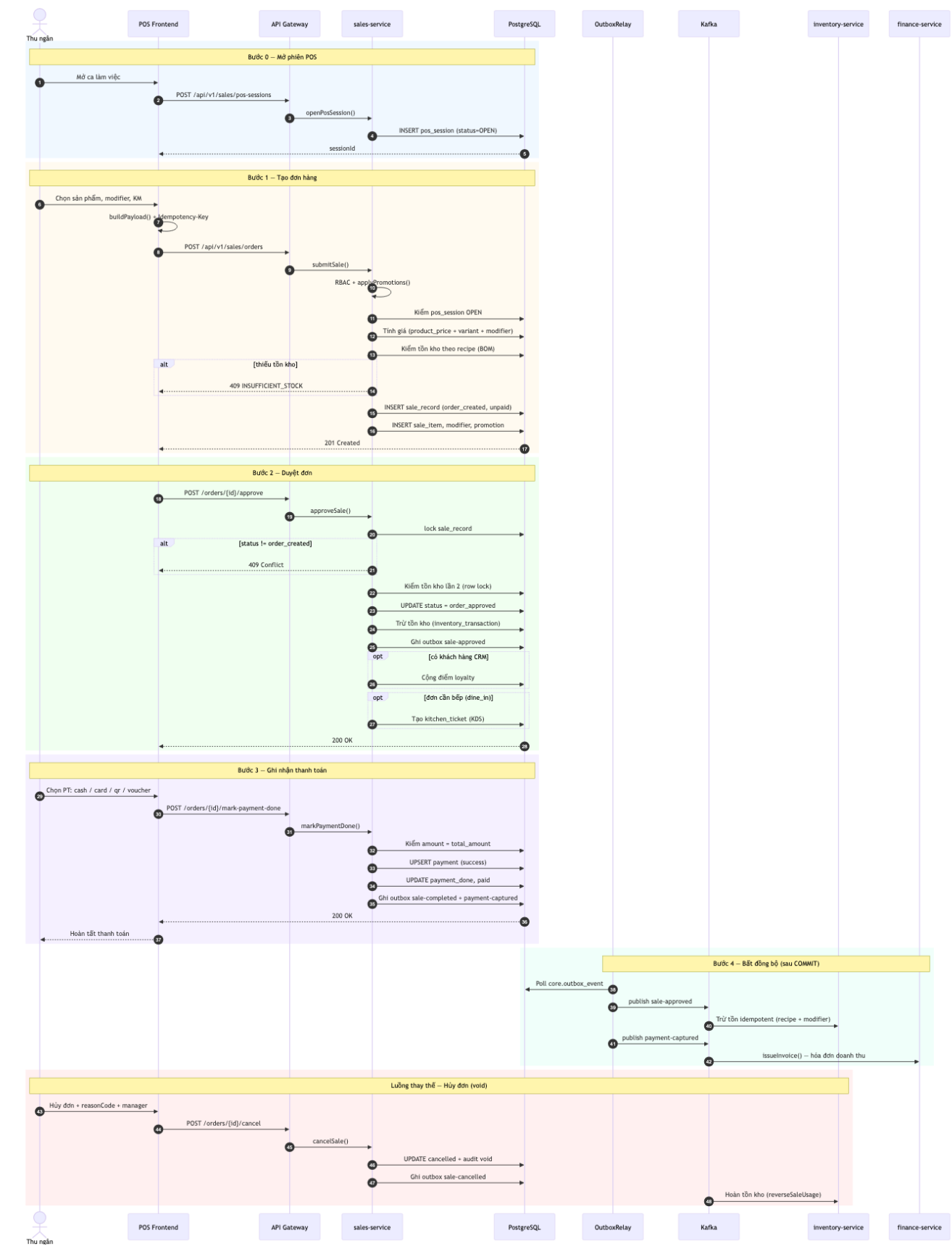


Luong ban hang POS (Order-to-Cash)

Luong nghiệp vụ dựa dựa từ source code FERN: mở phiên POS, tạo đơn, duyệt đơn (trú tồn kho), ghi nhận thanh toán, sự kiện bất đồng bộ (inventory, finance), và hủy đơn (void).

Sequence diagram



Bang tham chieu

1. Trang thai don hang

Buoc	API	status	payment_status
0. Mo phien	POST /api/v1/sales/pos-sessions	—	—
1. Tao don	POST /api/v1/sales/orders	order_created	unpaid
2. Duyet don	POST /api/v1/sales/orders/{id}/approve	order_approved	unpaid
3. Thanh toan	POST /api/v1/sales/orders/{id}/mark-payment-done	payment_done	paid
Huy don	POST /api/v1/sales/orders/{id}/cancel	cancelled	unpaid

2. Su kien Kafka (OutboxRelay)

Topic	Consumer	Tac dung
fern.sales.sale-approved	inventory-service	Tru ton kho idempotent (recipe + modifier)
fern.sales.sale-completed	audit / ClickHouse	Ghi nhan doanh thu hoan tat
fern.sales.payment-captured	finance-service	issueInvoice() - hoa don doanh thu
fern.sales.sale-cancelled	inventory-service	Hoan ton kho (reverseSaleUsage)
fern.sales.inventory-oversell	monitoring	Canh bao ban vuot ton (offline sync)

3. Thanh phan tham gia

Thanh phan	Vai tro
Thu ngan / POS Frontend	UI: chon SP, duyet, thanh toan; gui Idempotency-Key
API Gateway	Xac thuc JWT, routing toi sales-service
sales-service	SalesController, SalesService, SalesRepository
PostgreSQL (core.*)	sale_record, sale_item, payment, outbox_event, inventory_transaction
OutboxRelay + Kafka	Publish su kien sau COMMIT (transactional outbox)
inventory-service	Xu ly sale-approved / sale-cancelled
finance-service	Xu ly payment-captured, phat hanh hoa don

Ghi chu

- Tao don khong chap nhan payment trong body; thanh toan tach o buoc mark-payment-done.
- Ton kho duoc kiem 2 lan: khi tao don va khi duyet (row lock).
- Don dine_in co the tao kitchen_ticket (KDS); don co customer_id tu dong cong loyalty.
- Source: SalesController.java, SalesService.java, SalesRepository.java, use-submit-order.ts